Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")
```

Load dataset

```
df = pd.read_csv("/content/WA_Fn-UseC_-Telco-Customer-Churn.csv")
```

```
df.shape
```

```
(7043, 21)
```

```
df.columns
```

```
Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
       'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
       'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
       'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
      dtype='object')
```

Dispaly Data

```
df.head()
```

erID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneServic
590-VEG	Female	0	Yes	No	1	N
575-VDE	Male	0	No	No	34	Ye
668-YBK	Male	0	No	No	2	Ye
795-DCW	Male	0	No	No	45	N
237-QITU	Female	0	No	No	2	Ye

columns

Data Cleaning

```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
```

Drop missing values

```
df.dropna(inplace=True)
df.drop('customerID', axis=1, inplace=True)
```

Encode Target Variable

```
df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0})
```

Encode Categorical Features

```
df_encoded = pd.get_dummies(df, drop_first=True)
```

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion
```

Split Data

```
X = df_encoded.drop('Churn', axis=1)
y = df_encoded['Churn']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Train Model (Random Forest)

```
model = RandomForestClassifier(n_estimators=100, random_state=42)

model.fit(X_train, y_train)
```

```
RandomForestClassifier
RandomForestClassifier(random_state=42)
```

Predictions

```
y_pred = model.predict(X_test)
```

Model Evaluation

```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.7853589196872779
```

Classification

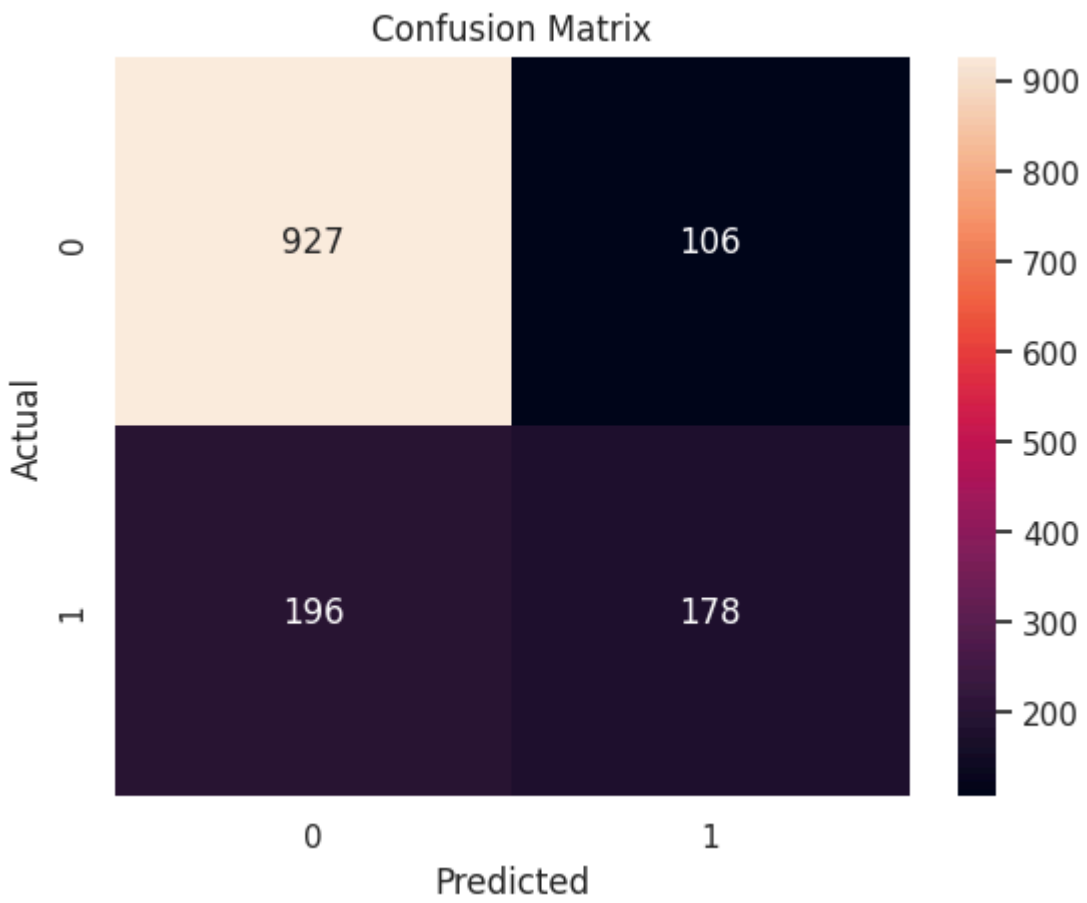
```
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.83      0.90      0.86      1033
           1       0.63      0.48      0.54       374

    accuracy                           0.79      1407
   macro avg       0.73      0.69      0.70      1407
```

```
weighted avg       0.77      0.79      0.78      1407
```

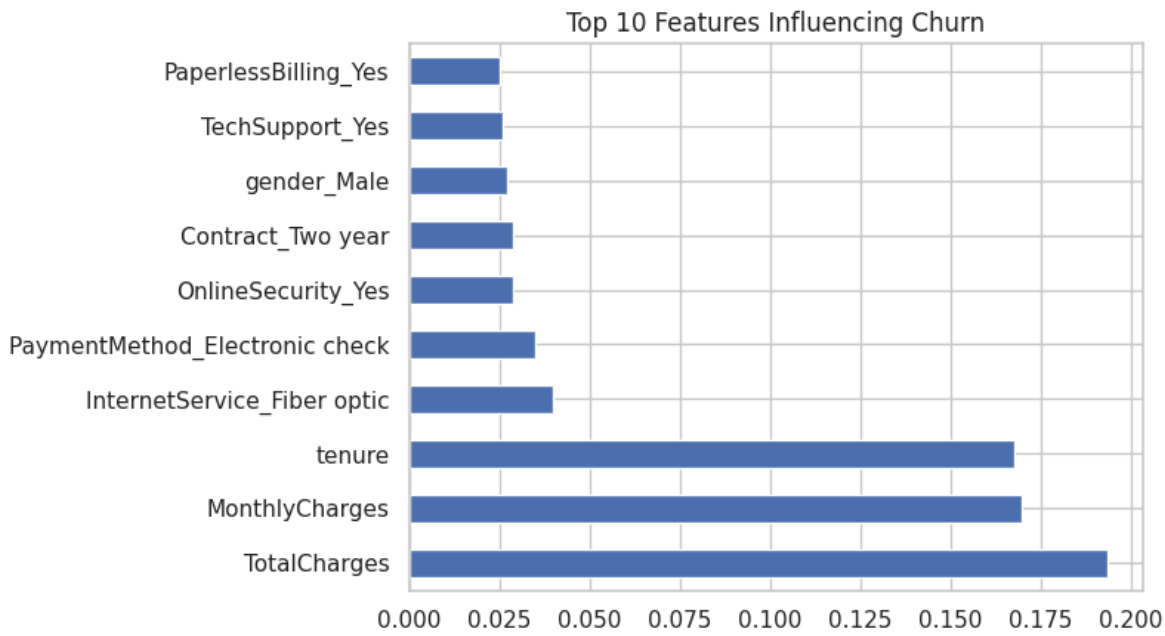
Confusion Matrix

```
cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



Feature Importance (Key Churn Drivers)

```
importance = pd.Series(model.feature_importances_, index=X.columns)

top_features = importance.sort_values(ascending=False).head(10)

top_features.plot(kind='barh')
plt.title("Top 10 Features Influencing Churn")
plt.show()
```



Sample Prediction

```
sample = X_test.iloc[0].values.reshape(1, -1)

prediction = model.predict(sample)

print("Predicted Churn:", "Yes" if prediction[0] == 1 else "No")
```

```
Predicted Churn: No
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: Use
  warnings.warn(
```